

Countable Borel Relations, Invariant Measures, and Graphings

Equivalence Relation Preliminaries

First, we collect some definitions.

Definition:

- A relation R on a set X is a set of ordered pairs $R \subseteq X^2$. We will write xRy to denote $(x, y) \in R$. We also define the *sections*

$$R_x = \{y : (x, y) \in R\}$$

$$R^y = \{x : (x, y) \in R\}$$

and the inverse relation

$$R^{-1} = \{(y, x) : (x, y) \in R\}.$$

- If X is equipped with a Borel structure, then the relation R (i.e., equivalence relations, partial orders, total orders, etc.) is called Borel if $R \subseteq X^2$ is Borel.
- We will let $\text{dom}(f)$, $\text{ran}(f)$, and $\text{graph}(f)$ denote the domain, range, and graph of f respectively.
- A graph G with vertex set X is a non-reflexive, symmetric relation on X . The neighbors of $x \in X$ in the graph G are the set of all $y \in X$ such that $(x, y) \in G$. The cardinality of the set of neighbors of x is called the degree of x , denoted $d_G(x)$.
- A path in G is a finite sequence of vertices $(x_0, x_1, \dots, x_n)$ where $(x_i, x_{i+1}) \in G$, and $x_i \neq x_j$ whenever $i \neq j$ except for possibly $x_n = x_0$. The equivalence classes of the relation xEy if and only if there exists a path from x to y defines the connected components of the graph. If a graph has one connected component, we say the graph is connected.
- If $x_n = x_0$ with $n \geq 3$, then we call the path a cycle. A graph that has no cycles is called acyclic, and a connected acyclic graph is called a tree. A rooted tree is a tree with a distinguished vertex x_0 called its root.
- If X is a standard Borel space and E is a Borel equivalence relation on X , then we say the relation is countable if $[x]_E$ is countable for each $x \in X$.
- If $\Gamma \curvearrowright X$ is a countable group acting on X such that Γ preserves the Borel structure of X , then we define the orbit equivalence relation to be $xE_\Gamma^X y$ if and only if there exists g such that $g \cdot x = y$.
- The orbit space X/Γ is defined to be the quotient X/E_Γ^X .
- If E is a countable Borel relation, then the *full group* of E , denoted $[E]$, is the set of Borel automorphisms whose graphs are contained in E . The set of partial Borel automorphisms is denoted $[[E]]$; that is,

$$[[E]] = \{\phi : A \rightarrow B : \phi \text{ is a bijection, } A, B \subseteq X \text{ are Borel, } \text{graph}(\phi) \subseteq E\}.$$

- We call a countable equivalence relation E on X *aperiodic* if every equivalence class $[x]_E$ is infinite.
- Given Borel equivalence relations E and F on X and Y respectively, we say E is *Borel reducible* to F , denoted $E \leq_B F$, if there is a Borel map $f : X \rightarrow Y$ such that xEy if and only if $f(x)Ff(y)$.

Example: If X^Γ denote the space of functions of Γ onto X , then the left shift action of Γ on X^Γ is defined by $\gamma \cdot p(\delta) = p(\gamma^{-1}\delta)$.

If G is a standard Borel group and $\Gamma \subseteq G$ is a countable discrete subgroup (also known as a lattice), then the action of Γ onto G by left translation, taking $\gamma \cdot g = \gamma g$, yields the quotient space G/Γ given by the set of *right* cosets of Γ .

Theorem (Feldman–Moore): Let E be a countable Borel equivalence relation on X . Then, there is a countable group Γ and a Borel action of Γ on X such that $E_\Gamma^X = E$.

Proposition (Marker Lemma): Let E be an aperiodic countable Borel equivalence relation on X . Then, there is a sequence $\{S_n\}_{n \geq 1}$ of Borel sets $S_n \subseteq X$ such that

- (i) $S_0 \supseteq S_1 \supseteq S_2 \supseteq \dots$;
- (ii) $\bigcap_{n \geq 1} S_n = \emptyset$;
- (iii) every S_n is a complete section for E (i.e., it has nonempty intersection with each equivalence class of E).

Such a sequence is called a *vanishing sequence of markers*.

Proof. Via some results from descriptive set theory, we may assume that the equivalence relation E lives on the set $X = 2^\mathbb{N}$. Then, we let $s_n(x)$ be the lexicographically smallest $s \in 2^n$ such that $|[x]_E \cap \mathcal{N}_s| = \infty$, where $\mathcal{N}_s$ denotes the cylinder set in X corresponding to s .

Define $x \in A_n$ if and only if $x|_n = s_n(x)$. Then, we have that $\{A_n\}_{n \geq 1}$ is a decreasing sequence of Borel sets which intersects each class of E at infinitely many points. Then, $A = \bigcap_{n \in \mathbb{N}} A_n$ intersects each class of E in at most one point, so that $B_n = A_n \setminus A$ is the vanishing sequence of markers as desired. $\square$

Invariant Measures

Definition: If E is a countable Borel equivalence relation on X , then we say μ is E -invariant if for all $f \in [E]$, we have $f_*\mu = \mu$, where $f_*\mu$ is the pushforward measure, defined by

$$f_*\mu(A) = \mu(f^{-1}(A)).$$

Proposition: The following are equivalent:

- (a) μ is E -invariant;
- (b) μ is Γ -invariant for any countable group Γ such that $E = E_\Gamma^X$;
- (c) μ is Γ -invariant for some Γ such that $E = E_\Gamma^X$;
- (d) for all $\phi \in [[E]]$, we have $\mu(\text{dom}(\phi)) = \mu(\text{ran}(\phi))$.

If μ is invariant under E , then we may define a measure $\bar{\mu}$ on E by taking

$$\bar{\mu}(A) = \int |A_x| d\mu(x),$$

where again,

$$A_x = \{y : (x, y) \in A\}.$$

It is also possible to define

$$\bar{\mu}'(A) = \int |A^y| d\mu(y),$$

but since μ is invariant, $\bar{\mu} = \bar{\mu}'$. This follows from the fact that

$$E = \bigcup_{i \in \mathbb{N}} \text{graph}(g_i)$$

for some collection of relation-preserving automorphisms $g_i \in [E]$. If $A \subseteq E$ is any Borel subset, then

$$A = \bigcup_{i \in \mathbb{N}} (\text{graph}(g_i) \cap A),$$

so there are $f_i \in [[E]]$ such that $\text{graph}(f_i) = \text{graph}(g_i) \cap A$. Therefore, we may write any $A \subseteq E$ as a countable disjoint union of graphs of functions $f \in [[E]]$. Therefore, it suffices to show that $\bar{\mu}(\text{graph}(f)) = \bar{\mu}'(\text{graph}(f))$ for any $f \in [[E]]$. If $C = \text{dom}(f)$ and $D = \text{ran}(f)$, then $\bar{\mu}(\text{graph}(f)) = \mu(C)$ and $\bar{\mu}'(\text{graph}(f)) = \mu(D)$, so by the definition of $[[E]]$, we are done.

Now, we may define an equivalence relation $\widetilde{E}$ on E by taking $(x, y)E(z, w)$ if and only if xEz , or equivalently, $xEyEzEw$.

Proposition: The invariant measure M on E is $\widetilde{E}$ -invariant.

Proof. Let Γ act on X such that $E = E_\Gamma^X$. Then, if we define the action $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$, then we have that $E_{\Gamma^2}^E = \widetilde{E}$, so it suffices to show that the action preserves M .

For this, if $A \subseteq E$ is Borel, we have

$$M((g, h) \cdot A) = \int \left| \{(g \cdot x, h \cdot y) : (x, y) \in A\} \right| d\mu(x).$$

Setting $B = (g, h) \cdot A$, then we have $y \in B_x$ if and only if $(x, y) \in B$, which holds if and only if $(g^{-1} \cdot x, h^{-1} \cdot y) \in A$, which holds if and only if $h^{-1} \cdot y \in A_{g^{-1} \cdot x}$. Therefore, $B_x = h \cdot A_{g^{-1} \cdot x}$, so that

$$\begin{aligned} M((g, h) \cdot A) &= \int |A_{g^{-1} \cdot x}| d\mu(x) \\ &= \int |A_x| d\mu(x) \\ &= M(A), \end{aligned}$$

where we use the fact that μ is invariant under Γ . □

References

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